The ethics of generative AI in software engineering

Edgar Jose Donoso Mansilla
e.donosomansilla@digipen.edu
DigiPen Institute of Technology
USA

ACM Reference Format:
Edgar Jose Donoso Mansilla. 2026. The ethics of generative AI in software engineering. In . ACM, New York, NY, USA, 4 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

Software engineering is a field that has developed most of today's infrastructure in some way or another, be it via the implementation of a medical device to the websites that allow us to connect the world. However, these projects need to manage several stakeholders, engineers and budgets to make things come to reality. With the improvements in generative AI (GenAI for short), there has been a shift towards trying to incorporate this new tool into the software development pipeline. These attempts are not small companies but instead industry leaders such as Microsoft. [13]

Given the workings of generative AI, there are several key concerns: The displacement of people for compute centers, the reliability of the software generated and the dilution of understanding the software developed. This essay will try to answer the question:

"Is it ethical for software engineers to use modern generative AI applications to generate code that is used, in whole or in part, for commercial applications?"

Conference'17, Washington, DC, USA

ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

2 Core Issues

As discussed in the introduction, there are several concerns with the use of generative AI in software development:

2.1 Software reliability

One of the major issues that stems from using GenAI is the fact that GenAI doesn't *reason* like humans do given its answer is generated via approximating using other good answers in the source. Programming is a task that requires discrete reasoning as the steps will directly relate to the next one in the solution [4]. This causes several issues when it comes down to the reliability of the software being written with the help of GenAI [9]:

- Mistakes might be missed when generating large amounts of code.
- Engineers can't guarantee where the design patters that were used to generate the system.
- Maintaining the systems will become more difficult over time as they were mainly written by GenAI as opposed to a human who had a stream of logic that was followed when designing the system.

These issues are more prominent when the software with generated code has critical importance such as a pacemaker, airplane cruising modes, etc. This issue compounds with GenAI's tendency to "hallucinate" an answer as its model might have biases and over fitting which make the math converge to a wrong answer [8].

2.2 Dilution of knowledge

Another major issue with GenAI for software engineering is the fact that they reduce the spaces where less experienced engineers build the knowledge to move forward. As stated before, GenAI doesn't reason and this is a similar issue with less experienced engineers, however, the key difference is that human engineers can improve with time and build up the discrete reasoning which allows them to construct correct programs that they haven't seen before. [12]

2.3 Displacement of people

This is a more general issue with the usage of GenAI but applies heavily to expensive tasks like generating the code for a complex program. The infrastructure that GenAI needs is costly in several resources [11] & [14]:

- About 22% power of all households in the US.
- "Chan School of Public Health found that the carbon intensity of electricity used by data centers was 48% higher than the US average."
- "Large data centers can consume up to 5 million gallons per day, equivalent to the water use of a town populated by 10,000 to 50,000 people."

Not only are data centers resource expensive, they also need a lot of space. This has caused local communities to be affected by the pollution and price increase of utilities. Not only that but there are cases where people lose the green areas of their community to a company who was able to buy out the land [3].

3 Stakeholders

With the core issues of GenAI out of the way, these are a possible classification of the different stakeholders in the situation (in order of least to most benefited):

(1) General population in developing countries
(2) General population in developed countries
(3) Software engineers
(4) Companies that use generative AI
(5) Companies that make generative AI tools

4 Current state of generative AI

With understanding of the costs of this technology it is also important to acknowledge its usage. By understanding the applications of the technology, it will be possible to weigh out the benefits and disadvantages of spending resources on this technology.

4.1 Vibe Coding

This no-code development experience differs from the professional world of software engineering where the tools are understood. This really doesn't have a major impact in the infrastructure of larger software but it does use all the same resources that other projects would use to improve their products [6].

4.2 Usage in research

There's been a shift in academia to use more GenAI to speed up different elements of the research pipeline [1]. A key project that will be mentioned in this essay is the usage to create the implementation for the ZOZO's (a Japanese clothing company) cloth physics solver [2].

4.3 Usage in open source

In open source projects there is a current issue where the lead contributors are being swarmed by pull requests with low quality AI generated code [10]. This puts a strain in projects' ability to survive as the leads are often people who volunteer and don't have the time to go through all the low quality code to pick out and accept contributions to the project.

4.4 Usage in private companies

Across different companies, there are several philosophies, some are encouraging their developers to use it and needing to strengthen their code reviews [5] and other companies avoid it as they feel it damages the quality of their products [7].

5 Technical Background

The technical section will focus on the following:

- The math of the models
- LLMs and how they work
- Code generation models and what extra features they implement to ensure correct code
- Why hallucinations occur

6 Ethical Analysis

6.1 Kantianism

Under Kantianism, the biggest ethical concerns are:

- Using code for training without the owner's conscent (GitHub Copilot)
- Displacing people to fit more compute centers
- Using more water to keep compute centers cool

6.2 Act Utilitarianism

In act utilitarianism the biggest contrast is whether all the negative effects on other humans helps maximize the happiness in other humans. This can be evaluated through the development of new technologies and research. By evaluating different sources it is possible to evaluate such cases.

6.3 Virtue Ethics

Under virtue ethics, the focus is the virtues of the stakeholders and how the decisions they make with this technology and how virtues either align or misalign with them.

6.4 Social Contract Theory

This section focuses on what our society expects out of software and how the usage of machine learning can breach the agreements and expectations that we have about the software that we use and how it is made.

7 Conclusion

8 Beyond the surface

References

[1] Jens Peter Andersen, Lise Degn, Rachel Fishberg, Ebbe K. Graversen, Evanthia Kalpazidou Schmidt Serge P.J.M. Horbach b, Jesper W. Schneider, and Mads P. Sørensen. [n. d.]. *Generative Artificial Intelligence (GenAI) in the research process – A survey of researchers' practices and perceptions.* https://cloud.google.com/discover/what-is-vibe-coding

[2] Ryoichi Ando. [n. d.]. *ZOZO's Contact Solver.* https://github.com/st-tech/ppf-contact-solver

[3] Pablo Jiménez Arandia. [n. d.]. *How We Investigated the Backyard of AI in Spain, Chile, and Mexico.* https://pulitzercenter.org/resource/how-we-investigated-backyard-ai-spain-chile-and-mexico

[4] R&A IT Strategy & Architecture. [n. d.]. *Generative AI 'reasoning models' don't reason, even if it seems they do.* https://ea.rna.nl/2025/02/28/generative-ai-reasoning-models-dont-reason-even-if-it-seems-they-do/

[5] AWS. [n. d.]. *AWS Responsible AI Policy.* https://aws.amazon.com/ai/responsible-ai/policy/

[6] Google Cloud. [n. d.]. *What is Vibe Coding?* https://cloud.google.com/discover/what-is-vibe-coding

[7] Peter Glagowski. [n. d.]. *Nintendo clarifies its stance on generative AI, says it has not had any contact with Japanese government over its usage.* https://nintendowire.com/news/2025/10/06/nintendo-clarifies-its-stance-on-generative-ai-says-it-has-not-had-any-contact-with-japanese-government-over-its-usage/

[8] IBM. [n. d.]. *What are hallucinations?* https://www.ibm.com/think/topics/ai-hallucinations

[9] Baskhad Idrisov and Tim Schlippe. [n. d.]. *Program Code Generation with Generative AIs.* doi:10.3390/a17020062

[10] Arjun Iyer. [n. d.]. *Open source maintainers are drowning in AI-generated pull requests. Enterprise teams are next.* https://thenewstack.io/ai-generated-code-crisis/

[11] James O'Donnellarchive and Casey Crownhart. [n. d.]. *We did the math on AI's energy footprint. Here's the story you haven't heard.* https://www.technologyreview.com/2025/05/20/1116327/ai-energy-usage-climate-footprint-big-tech/

[12] Phoebe Sajor. [n. d.]. *AI vs Gen Z: How AI has changed the career pathway for junior developers.* https://stackoverflow.blog/2025/12/26/ai-vs-gen-z/

[13] Jaime Teevan, Sonia Jaffe, Rebecca Janssen, Nancy Baym, Siân Lindley, Bahar Sarrafzadeh, Brent Hecht, Jenna Butler, Jake Hofman, and Sean Rintel. [n. d.]. *New Future of Work: AI is driving rapid change, uneven benefits.* https://www.microsoft.com/en-us/research/blog/new-future-of-work-ai-is-driving-rapid-change-uneven-benefits/

[14] Miguel Yañez-Barnuevo. [n. d.]. *Data Centers and Water Consumption.* https://www.eesi.org/articles/view/data-centers-and-water-consumption